

NORWAY HOUSE, MB, Canada

WMO: 714100

Lat: 53.960N

Lon: 97.840W

Elev: 734

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.3	-28.8	-41.0	0.5	-33.0	-36.7	0.7	-28.5	18.7	4.9	16.9	4.1	2.3	300	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	81.7	65.6	78.9	64.3	76.2	63.1	68.7	77.1	66.8	75.1	65.1	73.0	9.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	97.3	72.9	63.7	90.8	70.9	62.0	85.3	69.1	33.4	77.4	31.9	75.1	30.5	73.1	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.4	14.4	-39.6	86.8	5.4	2.9	-43.5	88.9	-46.7	90.6	-49.7	92.2	-53.6	94.3		
(5)			-39.8	72.0	5.4	2.8	-43.7	74.1	-46.8	75.7	-49.8	77.3	-53.7	79.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	-4.4	-0.3	13.9	31.8	46.2	58.6	64.7	62.0	51.6	37.1	18.2	1.6		
	DBStd	26.68	14.46	13.69	14.56	12.02	8.95	7.18	5.56	6.29	8.15	8.23	12.48	14.74		
	HDD50	7903	1687	1408	1120	551	183	13	0	1	77	408	955	1500		
	HDD65	12207	2152	1828	1585	996	584	212	75	135	406	864	1405	1965		
	CDD50	1301	0	0	0	4	64	271	454	373	126	9	0	0		
	CDD65	132	0	0	0	0	1	20	64	42	5	0	0	0		
	CDH74	1079	0	0	0	0	26	216	506	296	35	0	0	0		
	CDH80	166	0	0	0	0	2	35	82	43	4	0	0	0		
	WSAvg	6.7	6.1	6.3	6.9	7.4	7.7	7.0	6.5	6.7	6.9	7.1	6.5	5.9		
	PrecAvg	20.9	0.9	0.9	1.1	0.9	1.7	2.6	2.7	3.2	2.2	1.8	1.5	1.3		
	PrecMax	25.7	2.1	2.4	2.2	2.3	4.4	5.3	5.0	7.0	3.4	4.0	2.7	1.8		
(15) Precipitation	PrecMin	14.7	0.4	0.1	0.2	0.3	0.1	0.9	0.8	1.4	0.6	0.2	0.7	0.6		
	PrecStd	3.1	0.5	0.6	0.6	0.6	1.4	1.2	1.2	1.7	0.9	1.0	0.5	0.4		
	0.4%	DB	33.6	36.8	48.1	66.7	77.6	84.5	86.2	84.5	78.4	65.9	46.3	34.2		
	0.4%	MCWB	32.5	34.1	39.3	50.7	58.3	66.6	69.7	67.6	64.2	54.3	41.8	31.7		
	2%	DB	27.2	32.1	43.2	60.3	71.7	79.2	82.1	80.6	72.8	58.2	41.9	29.5		
(23) Mean Coincident Wet Bulb Temperatures	2%	MCWB	25.8	30.6	37.1	47.5	54.6	63.1	65.8	66.1	61.6	50.2	38.5	28.2		
	5%	DB	22.4	27.3	38.9	54.6	67.1	76.0	79.2	77.3	68.8	53.6	37.7	26.4		
	5%	MCWB	21.3	25.5	34.4	43.5	52.5	61.3	64.5	64.8	59.2	47.9	35.3	25.5		
	10%	DB	17.9	21.2	34.5	49.4	62.5	72.8	76.6	74.1	65.0	49.6	34.0	22.8		
	10%	MCWB	17.2	19.6	31.4	40.2	50.4	59.5	63.8	63.1	57.4	45.4	32.4	21.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.9	34.2	40.9	52.1	60.0	69.8	72.0	70.8	66.7	56.7	42.4	32.0		
	0.4%	MCDB	33.6	35.8	46.7	64.9	72.5	80.2	81.3	80.1	75.2	63.5	45.3	33.7		
	2%	WB	25.8	30.5	37.4	47.6	56.6	66.0	69.0	68.4	63.3	51.4	38.7	28.3		
	2%	MCDB	27.0	32.0	42.8	58.6	68.1	74.8	77.2	76.5	70.1	56.4	41.4	29.5		
	5%	WB	21.3	25.5	34.7	44.5	54.0	63.5	67.1	66.5	60.7	48.4	35.4	25.4		
(31) Mean Daily Temperature Range	5%	MCDB	22.2	27.3	38.6	53.5	64.2	72.4	75.6	74.4	67.4	53.0	37.6	26.3		
	10%	WB	17.1	19.7	31.7	40.9	51.4	61.3	65.3	64.7	58.1	45.8	32.6	22.0		
	10%	MCDB	17.8	21.2	34.5	48.5	60.9	69.8	73.5	71.7	64.0	49.4	33.9	22.8		
	5% DB	MDBR	16.6	20.6	22.5	21.5	20.6	20.0	18.8	17.8	16.7	12.6	13.2	15.2		
	5% DB	MCDBR	16.2	17.9	21.1	27.2	27.5	25.0	22.2	21.2	21.9	19.6	14.7	14.6		
(36) Clear-Sky Solar Irradiance	5% WB	MCWBR	15.5	16.0	15.6	16.4	13.9	11.8	9.6	9.9	12.2	12.7	11.7	13.7		
	5% WB	MCDBR	16.2	17.4	18.8	25.6	23.8	20.7	18.9	18.4	19.7	17.7	13.7	14.1		
	5% WB	MCWBR	15.5	15.7	14.0	15.8	13.1	11.0	9.8	10.1	12.1	12.0	11.2	13.3		
	taub		0.244	0.253	0.278	0.334	0.344	0.371	0.390	0.360	0.324	0.293	0.264	0.233		
	taud		2.264	2.300	2.330	2.372	2.441	2.375	2.359	2.468	2.525	2.568	2.360	2.253		
(42) All-Sky Solar Radiation	Ebn at Noon		234	274	290	284	287	278	270	273	271	259	225	214		
	Edh at Noon		18	25	30	34	33	36	36	30	25	19	16	15		
	RadAvg		271	607	1056	1554	1758	1816	1766	1441	1000	536	265	193		
(45)	RadStd		29	41	84	122	154	143	108	115	106	63	25	25		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page